

Number System

```
In [1]: bin(25) # Conversion number to binary number system
```

```
Out[1]: '0b11001'
```

```
In [2]: bin(30)
```

```
Out[2]: '0b11110'
```

```
In [3]: int(0b11001) # Binary number to number
```

```
Out[3]: 25
```

```
In [4]: int(0b11110)
```

```
Out[4]: 30
```

```
In [5]: bin(7)
```

```
Out[5]: '0b111'
```

```
In [6]: int(0b111)
```

```
Out[6]: 7
```

```
In [2]: 0b110
```

```
Out[2]: 6
```

```
In [3]: 0b101
```

```
Out[3]: 5
```

```
In [4]: print(0b110)
```

```
6
```

```
In [5]: print(0b101)
```

```
5
```

```
In [6]: print(bin(11))
```

```
0b1011
```

```
In [8]: print(0b1011)
```

```
11
```

```
In [9]: print(bin(12))  
print(bin(13))
```

```
0b1100  
0b1101
```

```
In [11]: 0b1101
```

```
Out[11]: 13
```

```
In [12]: 0b1100
```

```
Out[12]: 12
```

```
In [27]: bin(10)
```

```
Out[27]: '0b1010'
```

```
In [38]: print(bin(25))  
         print(bin(30))
```

```
0b11001
```

```
0b11110
```

```
In [37]: print(bin(35))  
         print(bin(40))
```

```
0b100011
```

```
0b101000
```

```
In [28]: 0b00111
```

```
Out[28]: 7
```

```
In [7]: oct(25)    # Conversion of number to octal number system
```

```
Out[7]: '0o31'
```

```
In [8]: int(0o31)  # Octal number to Number
```

```
Out[8]: 25
```

```
In [9]: hex(1)     # Conversion of Number to Hexadecimal number system
```

```
Out[9]: '0x1'
```

```
In [10]: hex(2)
```

```
Out[10]: '0x2'
```

```
In [34]: hex(8)
```

```
Out[34]: '0x8'
```

```
In [11]: hex(9)
```

```
Out[11]: '0x9'
```

```
In [35]: hex(10)
```

```
Out[35]: '0xa'
```

```
In [12]: hex(10)
```

Out[12]: '0xa'

In [36]: `hex(11)`

Out[36]: '0xb'

In [13]: `hex(16)`

Out[13]: '0x10'

In [14]: `int(0x10)`

Out[14]: 16

In [15]: `int(0x19)` *# Hexadecimal number to Number*

Out[15]: 25

In [16]: `hex(25)`

Out[16]: '0x19'

In [17]: `int(0xa)`

Out[17]: 10

In [18]: `int(0xb)`

Out[18]: 11

In [19]: `hex(256)`

Out[19]: '0x100'

In [20]: `int(0x19)`

Out[20]: 25

In [21]: `int(0x15)`

Out[21]: 21

Bitwise operator

1. Complement Operator

In [18]: `~12`

Out[18]: -13

In [19]: `~35`

Out[19]: -36

In [20]: ~18

Out[20]: -19

In [13]: ~46

Out[13]: -47

In [14]: ~54

Out[14]: -55

In [15]: ~10

Out[15]: -11

2. AND Operator

In [21]: 12 & 13

Out[21]: 12

In [16]: 1 & 0

Out[16]: 0

In [18]: 35 & 40

Out[18]: 32

3. OR Operator

In [23]: 12 | 13

Out[23]: 13

In [17]: 1 | 0

Out[17]: 1

In [19]: 35 | 40

Out[19]: 43

4. XOR Operator

In [24]: 12 ^ 13

Out[24]: 1

```
In [20]: 25 ^ 30
```

```
Out[20]: 7
```

5. Left Shift Operator

```
In [25]: 12 << 2
```

```
Out[25]: 48
```

```
In [26]: 12 << 1
```

```
Out[26]: 24
```

```
In [21]: 10 << 1
```

```
Out[21]: 20
```

```
In [22]: 10 << 2
```

```
Out[22]: 40
```

```
In [23]: 10 << 3
```

```
Out[23]: 80
```

6. Right Shift Operator

```
In [27]: 12 >> 1
```

```
Out[27]: 6
```

```
In [28]: 12 >> 2
```

```
Out[28]: 3
```

```
In [24]: 10 >> 1
```

```
Out[24]: 5
```

```
In [25]: 10 >> 2
```

```
Out[25]: 2
```

```
In [26]: 10 >> 3
```

```
Out[26]: 1
```

Swaping 2 Variables in Python

```
In [1]: a = 5  
b = 6
```

```
In [2]: a = b  
b = a
```

```
In [3]: print(a)  
print(b)
```

```
6  
6
```

```
In [1]: # In above scenario we lost the value 5
```

```
In [4]: a1, b1 = 7, 8  
print(a1)  
print(b1)
```

```
7  
8
```

```
In [5]: temp = a1  
a1 = b1  
b1 = temp
```

```
In [6]: # We're using third variable to swap the values  
print(a1)  
print(b1)
```

```
8  
7
```

```
In [7]: a2 = 5  
b2 = 6
```

```
In [8]: # Swap variable formulas without using 3rd variable  
a2 = a2 + b2 #5+6 = 11  
b2 = a2 - b2 #11-6 = 5  
a2 = a2 - b2 #11-5 = 6
```

```
In [10]: print(a2)  
print(b2)
```

```
6  
5
```

```
In [2]: print(0b101) # 101 is 3 bit  
print(0b110) # 110 is also 3 bit
```

```
5  
6
```

```
In [3]: # But when we use a2 + b2 then we get 11 that means we will get 4 bit which is 1  
print(bin(11))  
print(0b1011)
```

```
0b1011  
11
```

```
In [11]: # Swaping the variable value without using 3rd variable  
a3, b3 = 2,4
```

```
In [12]: a3, b3 = b3, a3
```

```
In [13]: print(a3)  
print(b3)
```

```
4  
2
```

```
In [31]: a4 = 10  
b4 = 15
```

```
In [32]: # There is other way to work using swap variable also which is XOR because it wi  
a4 = a4 ^ b4  
b4 = a4 ^ b4  
a4 = a4 ^ b4
```

```
In [33]: print(a4)  
print(b4)
```

```
15  
10
```

```
In [ ]:
```